Chapter 5

Thermochemistry and Thermodynamics

Learning Objectives

- System, Surroundings and Universe
- 1^{st} Law of Thermodynamics: internal energy, heat and work
- Enthalpy and Enthalpy change
  - Enthalpy of formation
  - Enthalpy of reaction
  - Bond enthalpy
- Hess’s Law
- Born-Haber Cycle

Controlling and Using Chemical Reactions

Seven questions:

Thermodynamics

- How much energy is absorbed/evolved by the reaction?
- Does the reaction proceed spontaneously, in a certain direction?
- How much useful work can be obtained from the reaction?
- What is the value of the equilibrium constant for the reaction?

Kinetics

- What is the rate of reaction?
- What factors affect the rate, and in what ways?
- What is the mechanism behind the reaction, and hence, what process limits the rate?

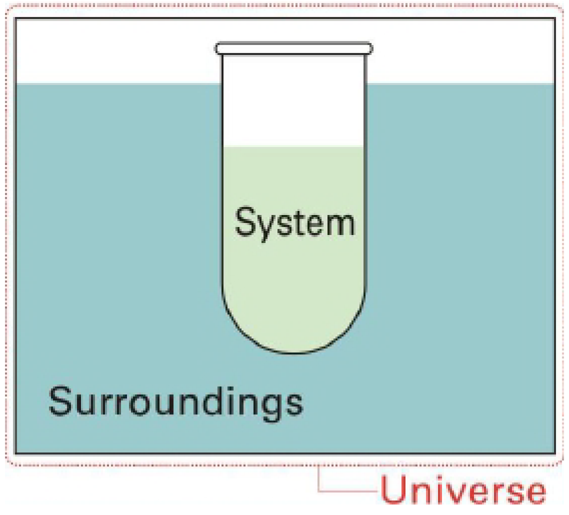
Part 1

1^{st} Law of Thermodynamics

System, Surroundings and Universe

- The system is the particular region of space (universe) that we are making measurements on.
- The surroundings would be everything else that is not the system.
- It follows then that

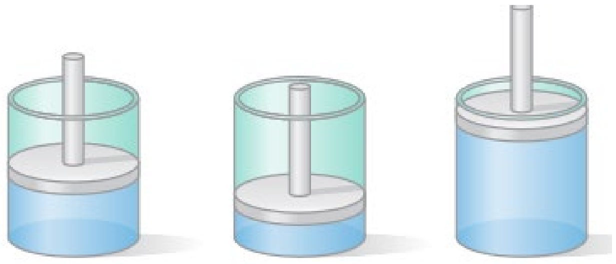
Universe = System + Surroundings



Typical examples of a system:

- The gas inside a vessel (may or may not include the vessel)
- The solution in a beaker
- A chemical reaction which include reactants and products

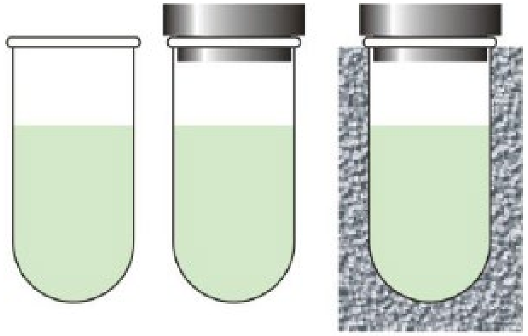
Classification of Systems

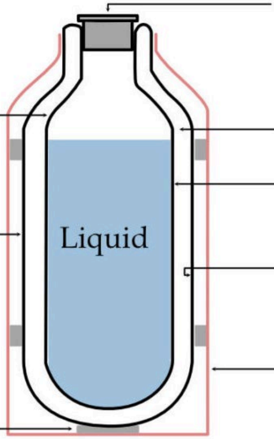


adiabatic – 绝热的

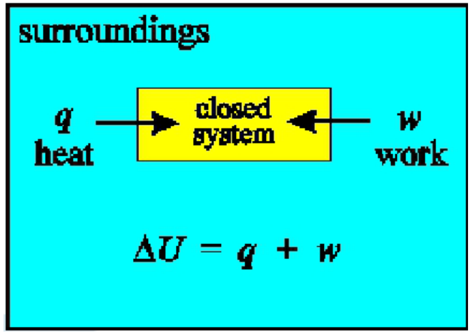
Class of System	Exchange of Matter	Exchange of Energy	Example
Isolated	No	No	• Dewar flask • Adiabatic, rigid gas cylinder
Closed	No	Yes	• A piston in a cylinder with a fixed amount of gas • A pressure cooker
Open	Yes	Yes	• Living system, e.g. animal, plant, human • A pot of boiling water

piston – 活塞





1^{st} Law of Thermodynamics



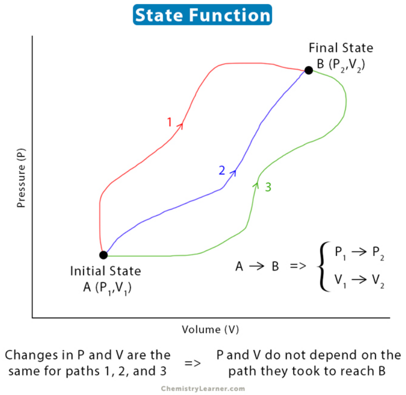
- $\Delta U = q + w$ [unit of each is joule, J]
- ΔU is the change of internal energy (total energy) of the system
- q is the heat added to the system (positive if heat is added to the system, negative if heat is removed from the system).
- w is the work added to the system (positive if work is done on the system, negative if work is done by the system on the surroundings).
  - Some other textbook use “work done by the system” – sign will be opposite

- The 1^{st} Law of Thermodynamics is indeed a special case of the more general “Law of Conservation of Energy”, which states that “energy can neither be created nor destroyed; rather, it can only be transformed or transferred from one form to another”.

- 能量守恒定律
  - 机械能守恒定律
  - 热力学第一定律

Internal Energy, U

- Internal Energy (U): the sum of all energies of a system, including kinetic energy (motion, vibration, rotation of molecules), potential energy (e.g. energy due to intermolecular attraction/repulsion), chemical energy (energy in chemical bonds), nuclear energy …
- Absolute value of U cannot be measured
- ΔU can be measured
- Change of internal energy (ΔU) may be a result of
  - heat passed into or out of the system;
  - work done on or by the system;
  - mass entered or left the system (not discussed in this course).

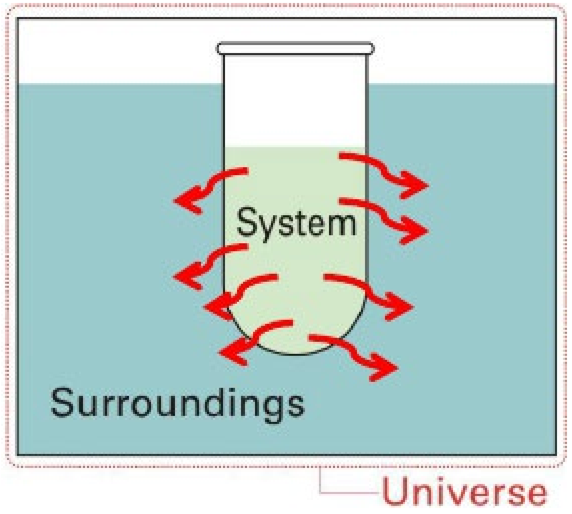


U is a state function

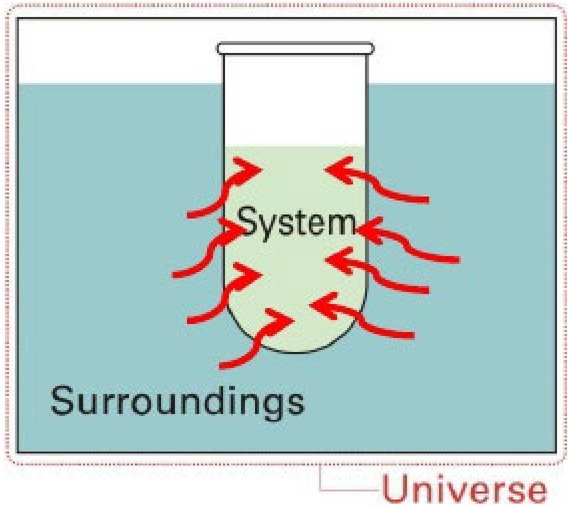
- All functions which refer to a system may be categorized as eit "*state functions*" or "*path functions*".
- A state function (e.g. internal energy) is one whose change (from initial to final) is independent of the route taken.
  - Once the state is fixed, the state function will have the same value, regardless which route it is taken to reach such state function
  - State function include: U, H, S, G, p, V, T (gravitational potential energy)
- Conversely, if the change in a function is dependent on the route taken, then the function is known as a path function or property (e.g. work done).
  - Path function include: q, w
- For solid, liquid and ideal gas, U is only a function of T and n → U is constant for isothermal process → $\Delta U_{isothermal} = 0$

Heat

- Adiabatic: if there is NO heat exchange between the system and the surrounding, the process is called “adiabatic”. Typically, the wall of the vessel is an adiabatic wall, insulating the heat transfer.



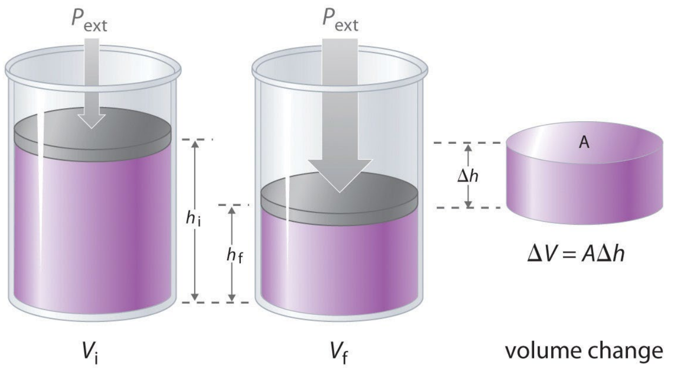
- Diabatic: if there is heat exchange between the system and the surrounding, the process is called “diabatic”. The wall is said to be thermally conductive, non-insulating or diathermic.
  - Endothermic: heat is transferred into the system
  - Exothermic: heat is transferred out of the system

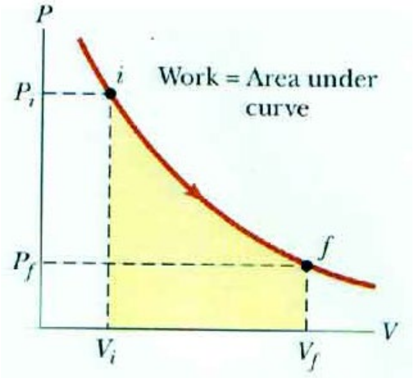


- Heat can be calculated by: $q = mc\Delta T$
  - c is the specific heat capacity of a substance. $C(H_2O) = 4.18\ J\ g^{-1}\ K^{-1}$

Work

- Work: pressure-volume work, gravitational work, elastic work electrical work …
- In thermodynamics, the most common work is pressure-volume work (p-V work)
    - In electrochemistry, the focus becomes electrical work
- Work with constant pressure: $w = -p_{ext}\Delta V = -p_{ext}(V_f - V_i)$
- Work with changing pressure: $-\int_{V_i}^{V_f} p_{ext}\,dV$
- Pay special attention to the sign: + or –
- p is p_{ext}, not p of the gas

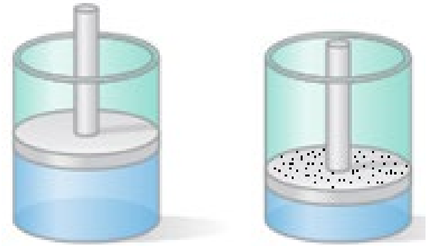




If external pressure is 0 → work is 0 → free expansion

Exercise: work in reversible isothermal process

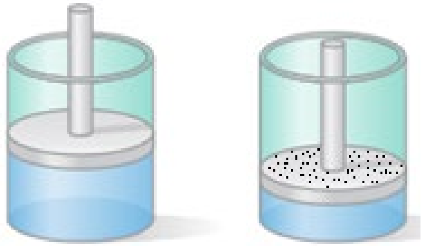
- A cylinder with frictionless and weightless piston contains ideal gas at T and 1 bar external pressure. Volume of gas is V_1.
- Sand grain is slowly added on top of the piston, one at a time. The mass of each sand grain is so small that the change of the gas can be considered as reversible. After the addition of the sand grains, the final volume of the gas $V_2 = 0.5V_1$.
- The process is isothermal, i.e. the temperature of the gas is constant.
- Calculate the initial pressure p_1 and the final pressure of the gas p_2.



- What is the ΔU of the process?

- Determine the sign of work (w) and heat (q) that is added to the system.

- At any instance, the internal pressure of the system and the external pressure are equal, because the piston is weightless and frictionless.
- Express the external pressure of the system in terms of n, V and T.
- Calculate the work and heat added to the system using w = $-\int_{V_i}^{V_f} p_{ext} dV$.



p_1, V_1, T $\quad$ p_2, V_2, T

Part 2

Enthalpy and Enthalpy Change

Enthalpy H and Heat Capacity

More chemical processes are under constant p than constant V
→ ΔH is more often used than ΔU

- Enthalpy H, is another thermodynamic state function
- It is defined as: $H = U + pV$ [note that each term is a state function]
- But why it is defined this way???
- Many chemical processes occur at constant (external) pressure (most often 1 atm), hence: $\Delta U = q + w = q_p - p\Delta V$
  - q_p is the heat of reaction at constant pressure, which can be experimentally measured
  - As a comparison, q_v is the heat of reaction at constant volume and $q_v = \Delta U$
- Therefore: $q_p = \Delta U + p\Delta V = (U_f - U_i) + p(V_f - V_i) = (U_f + pV_f) - (U_i + pV_i) = H_f - H_i = \Delta H$
- Numerically, for an isobaric process (constant p): $q_p = \Delta H$
  - But they are different: (H and) ΔH is a state function; q_p is a route function
  - Also, for a process with variable p, its $q \neq \Delta H$

Standard Enthalpy Change

- When all reactants and products are under “standard state”, the enthalpy change of the reaction is “standard enthalpy change”, ΔH^o
- Standard state (IUPAC definition)
  - Most stable form of a substance at a pressure of 1 bar (100 kPa, not 1 atm) and any specific temperature (most often 298.15 K, but can be any temp)
  - For elements with allotropes, the standard state is the most stable allotrope such as graphite for carbon and O_2 for oxygen, with the exception of white phosphorus P_4 (less stable than red phosphorus)
  - For solution, its standard concentration is 1 mol dm^{-3} (with infinite significant figures)
- The symbol for “standard state” is $\ominus$ in its superscript form. It is officially read as “plimssoll”, but more often as “nought”.
  - The $\ominus$ symbol is easy to write but difficult to type on computer, hence degree symbol or letter “o” in its superscript form is more often used.

List of Enthalpies

- **Enthalpy for Physical Transformations**
  - enthalpy of vaporization
  - enthalpy of condensation
  - enthalpy of fusion [for melting]
  - enthalpy of freezing
  - enthalpy of sublimation
- **Enthalpy for Chemical Transformations**
  - enthalpy of combustion
  - enthalpy of formation
  - enthalpy of reaction
  - enthalpy of neutralisation
  - enthalpy of solution

- enthalpy of atomisation
- bond dissociation enthalpy
- enthalpy of ionisation
- electron affinity/electron-gain enthalpy
- lattice enthalpy
- enthalpy of hydration

Enthalpy Change for Physical Transformation

(by convention, the name does not contain "change")

Standard Enthalpy of Vaporization

$H_2O\ (l) \rightarrow H_2O\ (g) \quad \Delta H^o_{vap} = +40.66\ kJ\ mol^{-1} \quad$ at 373 K

Standard Enthalpy of Condensation

$H_2O\ (g) \rightarrow H_2O\ (l) \quad \Delta H^o_{con} = -40.66\ kJ\ mol^{-1} \quad$ at 373 K

Standard Enthalpy of Fusion

$H_2O\ (s) \rightarrow H_2O\ (l) \quad \Delta H^o_{fus} = +6.01\ kJ\ mol^{-1} \quad$ at 273 K

Standard Enthalpy of Freezing

$H_2O\ (l) \rightarrow H_2O\ (s) \quad \Delta H^o_{freez} = -6.01\ kJ\ mol^{-1} \quad$ at 373 K

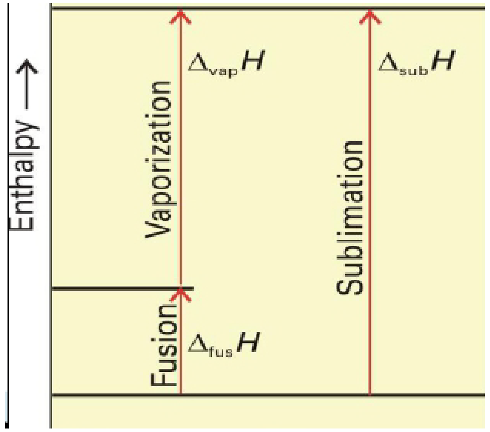
Standard Enthalpy of Sublimation

$I_2\ (s) \rightarrow I_2\ (g) \quad \Delta H^o_{sub} = +62.42\ kJ\ mol^{-1} \quad$ at 385 K

- How to remember the sign of the enthalpy?

The value is different by a "−" sign

The "+" sign is not a must, but it makes the value more clear



Exercise: Enthalpy of Sublimation

- Calculate the standard enthalpy of sublimation of ice at 253K.
- Average heat capacity (J g^{-1} K^{-1}): ice (-50 ^oC – 0 ^oC): 2.108; water (0 ^oC – 100 ^oC): 4.187 ; water vapor (-50 ^oC – 100 ^oC): 1.996

Enthalpy of Atomic and Molecular Change (1)

Standard Enthalpy of Atomization ΔH^o_{atom}

- The standard molar enthalpy change when 1 mole of the gaseous atoms are formed from its elements under standard conditions
- **$Na\ (s) \rightarrow Na\ (g)$** $\Delta H^o_{atom} = 107.3\ kJ\ mol^{-1}$ (298 K)
- **$\frac{1}{2}\ I_2\ (s) \rightarrow I\ (g)$** $\Delta H^o_{atom} = 106.0\ kJ\ mol^{-1}$ (298 K)
- **$\frac{1}{2}\ H_2\ (g) \rightarrow H\ (g)$** $\Delta H^o_{atom} = 216.0\ kJ\ mol^{-1}$ (298 K)

Note: the bonds broken are different

Standard Enthalpy of Ionization ΔH^o_{ion}

- The standard molar enthalpy change accompanying the removal of an electron from a gas-phase atom or ion. [Subsequent electron may be removed after the first.]
- **$Mg\ (g) \rightarrow Mg^+\ (g)$** $\Delta H^o_{ion,\ 1st} = 738\ kJ\ mol^{-1}$
- **$Mg^+\ (g) \rightarrow Mg^{2+}\ (g)$** $\Delta H^o_{ion,\ 2nd} = 1451\ kJ\ mol^{-1}$

Enthalpy of Atomic and Molecular Change (2)

Electron Affinity *(electron-gain enthalpy)* ΔH^o_{EA}

- The electron affinity of an element is the enthalpy change when 1 mole of gaseous atoms/ions accepts 1 mole of electrons to form 1 mole of anions.
- For most elements, the 1st EA ($X + e^- \rightarrow X^-$) is exothermic (except noble gases, nitrogen and a few metals)
- For all elements, the 2nd and subsequent EA (forming X^{2-}, …) is endothermic
- **$Cl\,(g) + e^- \rightarrow Cl^-\,(g)$** $\Delta H^o_{EA} = -364$ kJmol^{-1}
- **$O\,(g) + e^- \rightarrow O^-\,(g)$** $\Delta H^o_{EA} = -142$ kJmol^{-1}
- **$O^-\,(g) + e^- \rightarrow O^{2-}\,(g)$** $\Delta H^o_{EA} = +844$ kJ mol^{-1}

By convention, electron affinity is taking a positive sign for the exothermic 1st EA.

But in our course, we will keep the same sign convention as other enthalpies.

Enthalpy of Atomic and Molecular Change (3)

Lattice Energy *(lattice enthalpy)*

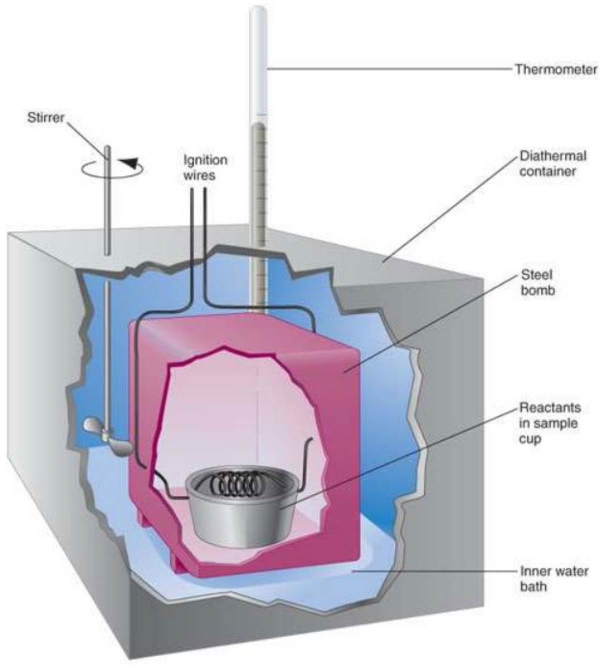
- The enthalpy change when one mole of the crystalline substance is formed from its gaseous ions, e.g. $Na^+(g) + Cl^-(g) \rightarrow NaCl(s)$
  - Some other books: enthalpy of the opposite process $NaCl(s) \rightarrow Na^+(g) + Cl^-(g)$
- **$Na^+ (g) + Cl^- (g) \rightarrow NaCl (s)$** $\Delta H^o_{lat} = -774\ kJmol^{-1}$
- Lattice energy is a measurement of the ionic bond strength
  - The higher the charge of the ions → the stronger the ionic bond
    - NaCl (-774) vs $MgCl_2$ (-2500); MgO (-3950) vs TiO_2 (-12150); $MgCl_2$ (-2500) vs MgO (-3950)
  - The larger the size of the ions → the weaker the ionic bond
    - LiF, NaF, KF: -1024, -904, -830
- Lattice energy cannot be directly measured by experiment (why?)
  - Use Born-Haber cycle to calculate (see Part 3)

Enthalpy of Chemical Change (1)

Standard Enthalpy of Combustion, ΔH_c^o or $\Delta_c H^o$

- when one mole of the pure substance is completely burned (oxidised, combusted) in excess oxygen under standard conditions.

- Enthalpies of combustion are commonly measured by using a *bomb calorimeter*, the heat is q_v.
- In order to obtain ΔH ($=q_p$):
  - $\Delta U = q+w \rightarrow$ @const. V, w=0 $\rightarrow \Delta U = q_v$
  - $\Delta H = \Delta U + p\Delta V = \Delta U + \Delta(nRT) = \Delta U + (\Delta n)RT = q_v + (\Delta n)RT$
  - Δn is the difference of gaseous molecules in the balanced equation. Liquid/solid/solution is not counted.



Temperature is important, because water may be in liquid / gas state depending on the temperature

Enthalpy of Chemical Change (2)

Standard Enthalpy of Formation, ΔH_f^o or $\Delta_f H^o$

- when one mole of the pure substance is formed from its elements under standard conditions.
- $H_2(g) + ½O_2\ (g) \rightarrow H_2O\ (l)$ $\Delta H_f^o = -285.5$ kJ mol^{-1}
  - This can be experimentally measured. In fact, it is the same as ΔH_c^o of H_2.
- $2C$ (s, graphite) $+ 3H_2(g) + ½O_2(g) \rightarrow C_2H_5OH\ (l)$ $\Delta H_f^o = -277.7$ kJ mol^{-1}
  - This cannot experimentally measured. It can only be calculated from other reactions by Hess's Law.
- By definition the standard molar enthalpy change of formation of an ***element*** in its standard state is ***zero***.
  - $N_2(g) \rightarrow N_2(g)$ ΔH_f^o must be zero. $3/2O_2(g) \rightarrow O_3\ (g)$ ΔH_f^o is not zero for ozone.

Enthalpy of Chemical Change (3)

- If ΔH_f^o of many substances are not experimentally available, why do scientists want to collate such data?
- ΔH_f^o is very useful! With a table of ΔH_f^o, scientists can calculate the enthalpy change of any chemical reaction, regardless whether such reactions can actually occur or not.
- Example : $C(s, graphite) + \frac{1}{2}O_2(g) \rightarrow CO(g)$
  - Experimentally, its ΔH_r^o cannot be measured, because i) with sufficient oxygen, the product is CO_2; ii) with insufficient oxygen, the product is $CO_2 + CO$ (maybe with unreacted C) but never pure CO.

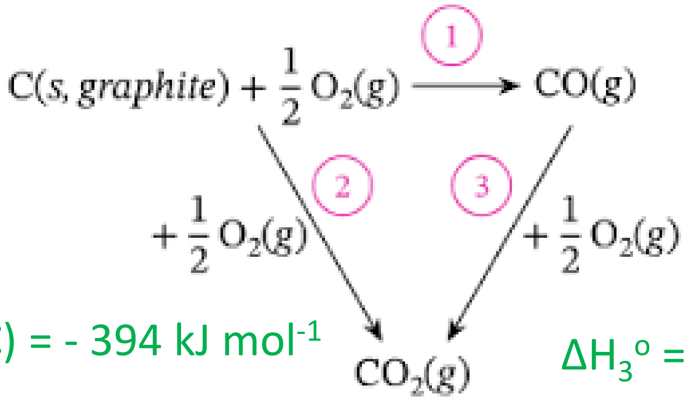
Part 3

Hess's Law and Born-Haber Cycle

Hess's Law

- The total enthalpy change during the complete course of a chemical reaction is independent of the sequence of steps taken - ΔH is path independent.
- If a chemical equation can be written as the sum of several other chemical equations, the enthalpy change of the first chemical equation equals the sum of the enthalpy changes of the other chemical equations.

ΔH_1^o = ???



$\Delta H_2^o = \Delta H_c^o(C)$ = - 394 kJ mol^{-1}

$\Delta H_3^o = \Delta H_c^o(CO)$ = - 283 kJ mol^{-1}

Sum of which two equations gives the 3rd equation?

(1) + (2) = (3) ?

(2) + (3) = (1) ?

(3) + (1) = (2) ?

Exercise: stability of graphite and diamond

- Which allotrope of carbon is more stable thermodynamically?
- Consider the reaction: C(s, diamond) → C(s, graphite)
- Calculate the ΔH_r^o by Hess's Law, given:

$$C\,(s,\ graphite) + O_2(g) \rightarrow CO_2(g) \qquad \Delta H_1^o = -393.51\ kJ\ mol^{-1}$$

$$C\,(s,\ diamond) + O_2(g) \rightarrow CO_2(g) \qquad \Delta H_2^o = -395.40\ kJ\ mol^{-1}$$

Exercise: $\Delta_f H^o$ of benzene and cyclohexane

- Most substances cannot be prepared directly from their elements, hence $\Delta_f H^o$ are seldom determined by direct measurement. Rather $\Delta_f H^o$ are calculated by Hess's Law from other ΔH^o, commonly $\Delta_c H^o$.
- Determine $\Delta_f H^o$ of benzene and cyclohexane, given $\Delta_c H^o$ of:
  - Benzene (C_6H_6, l): -3268 kJ mol^{-1}
  - Graphite (s): 393.5 kJ mol^{-1}

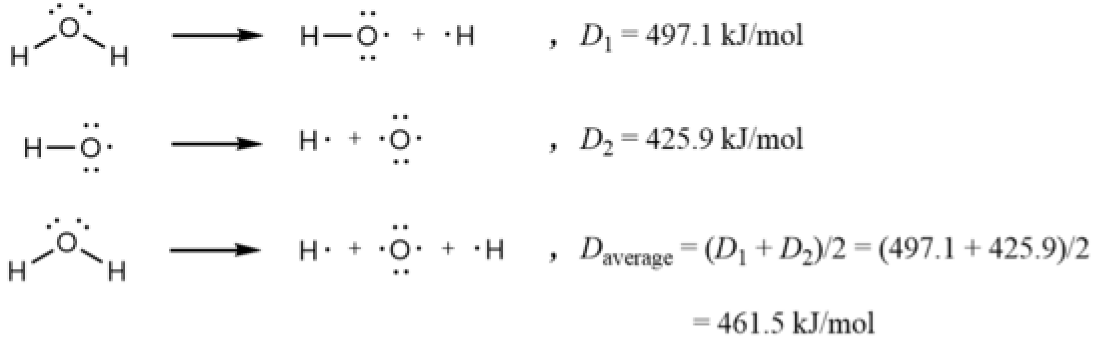
  Cyclohexane (C_6H_{12}, l): 3920 kJ mol^{-1}

  Hydrogen (g): -285.5kJ mol^{-1}

Bond Energy (1)

Bond Energy (Bond Enthalpy, Bond Dissociation Energy - BDE)

- The enthalpy of breaking a bond to form atoms (not ions).
  - HCl (g) → H (g) + Cl (g)   BDE(H—Cl) = +431 $kJmol^{-1}$
  - If there is more than one bond, then take the average value



Bond Energy (2)

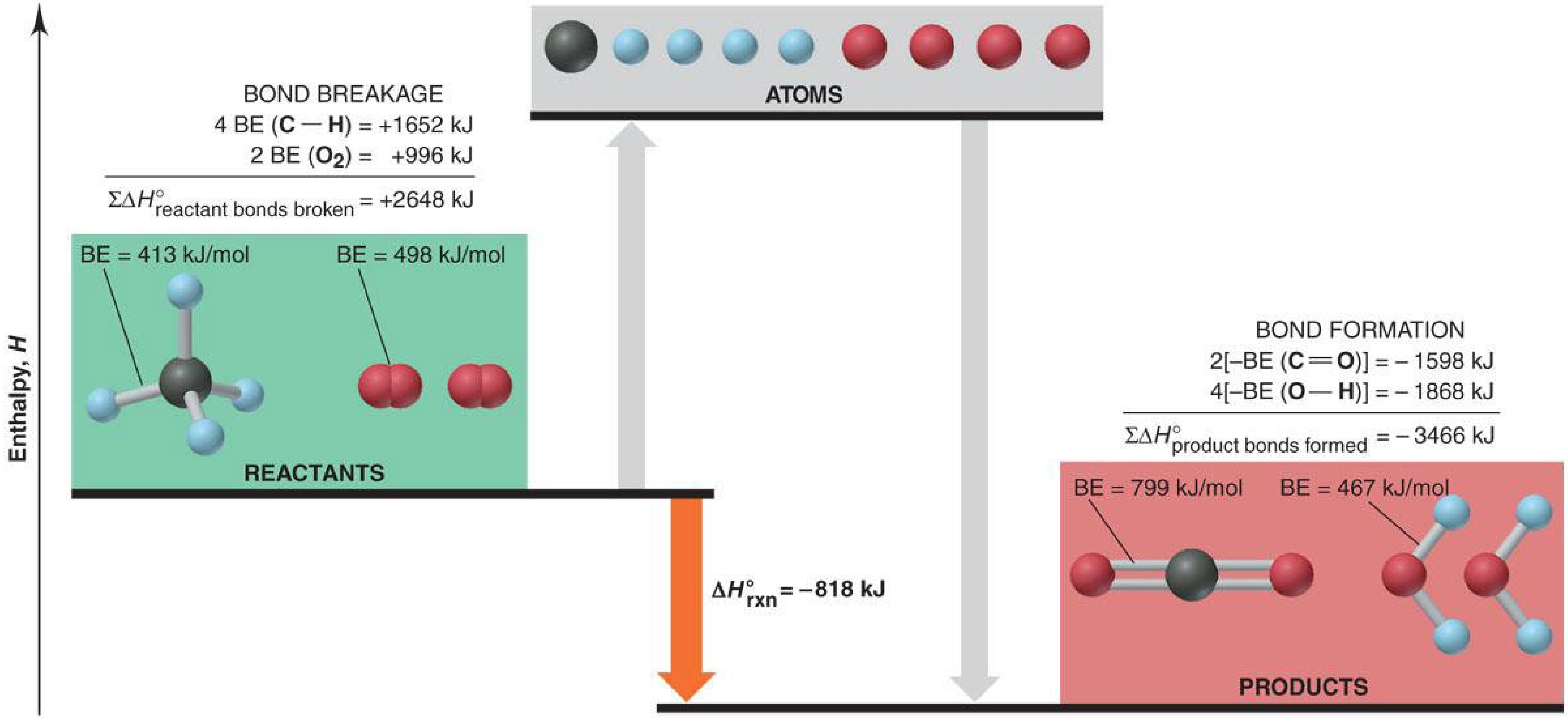
- In practice, average bond energy is tabulated and used.
  - Use C-O bond as an example, alcohol (CH_3-OH, CH_3CH_2-OH, …), phenol (C_6H_5-OH, …), ether (CH_3-O-CH_3, CH_3CH_2-O-CH_3, …), carboxylic acid (HCO-OH, CH_3CO-OH, …), ester (CH_3CO-OCH_3, CH_3CO-OCH_2CH_3, …) all have slightly different C-O bond energy
  - It would be too tedious (or even impossible) to list out each C-O bond energy
  - Since such C-O energies are quite close to each other, only their average value is tabulated.
  - Although accurate BDE gives accurate $\Delta_r H^o$, the average BDE is good enough for most applications.
- Hess's Law is likewise applied on BDE to give $\Delta_r H^o$.

Average Bond Energy Table

Table 9.2 Average Bond Energies (kJ/mol) and Bond Lengths (pm)

Bond	Energy	Length	Bond	Energy	Length	Bond	Energy	Length	Bond	Energy	Length
Single Bonds											
H—H	432	74	N—H	391	101	Si—H	323	148	S—H	347	134
H—F	565	92	N—N	160	146	Si—Si	226	234	S—S	266	204
H—Cl	427	127	N—P	209	177	Si—O	368	161	S—F	327	158
H—Br	363	141	N—O	201	144	Si—S	226	210	S—Cl	271	201
H—I	295	161	N—F	272	139	Si—F	565	156	S—Br	218	225
			N—Cl	200	191	Si—Cl	381	204	S—I	~170	234
C—H	413	109	N—Br	243	214	Si—Br	310	216			
C—C	347	154	N—I	159	222	Si—I	234	240	F—F	159	143
C—Si	301	186							F—Cl	193	166
C—N	305	147	O—H	467	96	P—H	320	142	F—Br	212	178
C—O	358	143	O—P	351	160	P—Si	213	227	F—I	263	187
C—P	264	187	O—O	204	148	P—P	200	221	Cl—Cl	243	199
C—S	259	181	O—S	265	151	P—F	490	156	Cl—Br	215	214
C—F	453	133	O—F	190	142	P—Cl	331	204	Cl—I	208	243
C—Cl	339	177	O—Cl	203	164	P—Br	272	222	Br—Br	193	228
C—Br	276	194	O—Br	234	172	P—I	184	243	Br—I	175	248
C—I	216	213	O—I	234	194				I—I	151	266
Multiple Bonds											
C=C	614	134	N=N	418	122	C C	839	121	N N	945	110
C=N	615	127	N=O	607	120	C N	891	115	N O	631	106
C=O	745 (799 in CO_2)	123	O_2	498	121	C O	1070	113			

Calculate the $\Delta_r H^o$ for combustion of methane by BDE



Calculation of $\Delta_r H^o$

- By $\Delta_f H^o$: $\Delta_r H^o = \Sigma \Delta_f H^o_{\textbf{products}} - \Sigma \Delta_f H^o_{\textbf{reactants}}$
  - Remember to multiply individual $\Delta_f H^o$ by the stoichiometric number
  - Pay attention to the sign: + or –
  - It's always products minus reactants, not the reverse!
  - Reaction involving ions can be calculated the same way, as long as $\Delta_f H^o$ of the ions are available
  - If you switch the reactants and the products, ΔH^o will have an opposite sign
- By BDE: $\Delta_r H^o = \Sigma$ BDE (bonds broken) – Σ BDE (bonds formed)
  - Again, remember to multiply by the stoichiometric number (of molecules and of the number of bonds within the molecule)
  - Commonly used for organic reactions, as $\Delta_f H^o$ of many organic compounds are not available

Born-Haber Cycle

- We have learnt the concept “Lattice Energy” in Chapter 3. We know the trend, but how about the accurate value?
- Lattice Energy cannot be experimentally measured (why?). It can only be calculated by Hess’s Law in the Born-Haber Cycle.

What is each of the energy in the cycle?

What is the LE of LiF?